

B.Tech, Mathematics-II
Mid-Term examination even semester 2023-24
22MAT102

Maximum Marks: 30

Time: 90 minutes

Date: 26 February, 2024

Each question carries equal marks.

1. ✓ Solve $(D^2 + 5D - 6)y = \sin x \sin 4x$, $D = \frac{d}{dx}$.

2. Write the Charpit's Auxiliary equations and solve the following PDEs:

(a) $p(1 + q) = qz$

(b) $p^2 + q^2 = x + y$

3. ✓ Consider $\Phi(u, v) = 0$, where u, v are known functions of x, y, z and Φ is an arbitrary function. Find the partial differential equation by eliminating the arbitrary function Φ .

4. ✓ Obtain a series solution of the differential equation $(1 - x^2)y'' - 4xy' + 2y = 0$ around the origin using the power series method. Find at least three non-zero terms in each linearly independent series solution.

5. ✓ Solve $(D^2 + 2DD' + 3D'^2)z = e^{x-2y}$, $D = \frac{\partial}{\partial x}$, $D' = \frac{\partial}{\partial y}$.

6. ✓ Find the general solution of the equation $x^3y''' + 5x^2y'' + 5xy' + y = x^2 + \ln x$, $x > 0$

2. ✓ Consider a random experiment of rolling two dice simultaneously. Find the expectation and variance of the modulus of the difference of the two numbers that come up. [3]

✓ (b) Suppose X and Y are two random variables. If $V[X] = 4$ and $V[4Y] = 32$, then find the variance of $Z = 2X - 8$ and $U = Y + 2$. [2]

✓ (c) Write the importance of Gaussian distribution in data science. Derive the variance of a random variable X that follows Gaussian distribution. [4]

→ (d) ✓ Define chain rule in case of conditional density function. [1]

3. (a) ✓ The joint density function of X and Y is given by

$$p(x, y) = \alpha(2x + y); \quad x = 0, 1, 2; \quad y = 0, 1, 2$$

Find (i) ✓ the value of α (ii) ✓ marginal distribution of X and Y (iii) ✓ Are X and Y are independent variables? (iv) conditional distribution of Y for $X = x$. [4]

✓ (b) ✓ What is meant by convex function? Determine whether the function $f(x) = \sqrt{1 - x^2}$ is convex or concave. [3]

✓ (c) ✓ How is entropy related to expectation? Calculate the entropy in bits for each of the following random variables: 42 [3]

✓ (i) ✓ Pixel values in an image whose possible grey values are all the integers from 0 to 7 with uniform probability.

✓ (ii) ✓ A population of persons classified by whether they are older, or not older, than the population's median age.

***** All the Best *****

Malaviya National Institute of Technology Jaipur
Department of Mathematics
B.Tech./Sem-II/Mid-Term Examination: 2024-25
22MAT151: Mathematics II

Date: Feb. 21, 2025

Max. Mark: 30

Time: 90 minutes

Remark: Calculator is not allowed.

1. (a) Find the ordinary points of the differential equation (1)

$$(x-1)(x+2)\frac{d^2y}{dx^2} + x^2\frac{dy}{dx} + xy = 0.$$

- (b) Obtain the solution of the differential equation: $\frac{d^4y}{dx^4} + 6\frac{d^2y}{dx^2} + 9y = 0.$ (2)

- (c) Find the solution of the differential equation: $x^2\left(\frac{\partial z}{\partial x}\right)^2 + y^2\left(\frac{\partial z}{\partial y}\right)^2 = z^2.$ (2)

2. If $y = y_1(x)$ be a non-trivial solution of the following differential equation, then find its general solution by **method of reduction of order**, (5)

$$a_0(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_2(x)y = 0, \quad a_0(x) \neq 0 \quad \text{on a given interval } I.$$

3. Using **method of variation of parameter**, solve the following differential equation (5)

$$x^2\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + 6y = x^2 \log x.$$

4. Obtain power series solution of the following differential equation (5)

$$\frac{d^2y}{dx^2} - x\frac{dy}{dx} + 2y = 0, \quad \text{about } x = 1.$$

5. Solve the following differential equation (5)

$$(x^2 - y^2 - yz)\frac{\partial z}{\partial x} + (x^2 - y^2 - zx)\frac{\partial z}{\partial y} - z(x - y) = 0.$$

6. Find the partial differential equation by eliminating arbitrary functions: (5)

$$z = x f(x+t) + g(x+t).$$

$\cdot 16 \times 24$
 $49 \left(\frac{1}{m^2} + \frac{1}{16m} + \frac{9}{m} \right)$

$\frac{1}{12}$
 $\frac{1}{12}$
 $\frac{1}{12}$

Malaviya National Institute of Technology Jaipur
Department of Mathematics

B.Tech./II sem/2025-26/MID TERM EXAM
22MAT151- Mathematics II

Time: 1 Hour 30 Minutes Max Marks:30 February 23, 2026

Note: Use of calculator is not allowed.

Q.1 (a) Let y_1, y_2 be two linear independent solutions of the equation $y'' + a(x)y' + b(x)y = 0$ on an open interval I . Show that there is no point $x_0 \in I$ at which both y_1, y_2 take extreme values. [3 Marks]

(b) Find a homogeneous differential equation for which the functions e^{2x} , $\sin x$, and $\cos x$ are solutions. [2 Marks]

Q.2 Solve the differential equation $y''' - 3y'' + y' - 3y = x^2$. [5 Marks]

Q.3 Solve the following differential equation

$$(x^2 + 1)y'' - 2xy' + 2y = 6(x^2 + 1)^2;$$

given that $y_1 = x$ is a solution of the corresponding homogeneous equation. [5 Marks]

Q.4 Find the power series solution about the point $x = 1$ of the differential equation

$$(1 + 2x)y'' + xy' + y = 0.$$

[5 Marks]

Q.5 In the following problems $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$.

(a) Find partial differential equation whose general solution is given by

$$\phi(x + yz, x^2 + y^2 - z^2) = 0,$$

where ϕ is an arbitrary function. [3 Marks]

(b) Solve $y^2(x + y)p + x^2(x + y)q = (x^2 + y^2)z$. [3 Marks]

(c) Solve $p^2y = q(z - qy)$. [4 Marks]

.....Question Paper Ends Here.....

you can

$$\frac{du}{y} = \frac{dy}{x}$$

$$\frac{du}{x+yn} = \frac{dy}{x+yn}$$

$$\frac{\partial(u,v)}{\partial(y,z)} \Rightarrow \begin{vmatrix} 3 & y \\ 2y & -2z \end{vmatrix} \Rightarrow -2z^2 - 2y^2$$

$$\frac{\partial(u,v)}{\partial(z,n)} \Rightarrow \begin{vmatrix} y & 1 \\ -2z & 2n \end{vmatrix} \Rightarrow 2ny + 2z$$

$$\frac{\partial(u,v)}{\partial(n,y)} \Rightarrow \begin{vmatrix} 1 & z \\ 2n & 2y \end{vmatrix} \Rightarrow 2y - 2nz$$